

## PROVIDENCE GREEN, RI, USA

WMO: 725070

Lat: 41.723N

Lon: 71.433W

Elev: 55

StdP: 14.67

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14765

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 8.1        | 12.8 | -10.0                         | 3.2 | 11.9 | -5.6 | 4.1 | 16.4 | 28.5                  | 39.1 | 25.8 | 36.6 | 10.0                  | 310  | 0.543 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 16.8                   | 90.0            | 73.5 | 86.8 | 72.1 | 84.0 | 70.6 | 76.6                | 85.4 | 75.1 | 82.2 | 73.6 | 79.9 | 11.7                 | 290  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 74.0                            | 126.9 | 80.5 | 72.7 | 121.7 | 78.9 | 71.5 | 116.7 | 77.6 | 39.9          | 85.7 | 38.5 | 82.2 | 37.1 | 80.1 | 81.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |       |            |       |       |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|-------|------------|-------|-------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |       | n=50 years |       |       |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max   | Min        | Max   |       |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)   | (o)        | (p)   |       |
| 24.1              | 20.5 | 18.5 | 2.4                        | 95.5 | 5.3                | 3.2 | -1.4                                               | 97.8 | -4.5       | 99.7 | -7.5       | 101.5 | -11.4      | 103.8 |       |
|                   |      |      | DB                         | 2.4  | 95.5               | 5.3 | 3.2                                                | -1.4 | 97.8       | -4.5 | 99.7       | -7.5  | 101.5      | -11.4 | 103.8 |
|                   |      |      | WB                         | 0.7  | 79.1               | 5.0 | 1.5                                                | -2.9 | 80.2       | -5.9 | 81.1       | -8.7  | 82.0       | -12.3 | 83.1  |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 52.2   | 30.0  | 32.0  | 38.8  | 49.3  | 58.8  | 68.0  | 74.3  | 72.9  | 65.8  | 54.6  | 44.5  | 35.8 |
|                                                                                  | DBStd       | 17.08  | 10.03 | 9.05  | 8.47  | 7.20  | 7.04  | 6.53  | 5.12  | 5.14  | 6.15  | 7.32  | 8.17  | 8.70 |
|                                                                                  | HDD50       | 2276   | 622   | 507   | 360   | 95    | 5     | 0     | 0     | 0     | 0     | 35    | 203   | 449  |
|                                                                                  | HDD65       | 5477   | 1085  | 925   | 812   | 472   | 221   | 43    | 2     | 3     | 64    | 330   | 614   | 906  |
|                                                                                  | CDD50       | 3072   | 2     | 2     | 13    | 75    | 277   | 541   | 754   | 709   | 474   | 179   | 39    | 7    |
|                                                                                  | CDD65       | 798    | 0     | 0     | 0     | 2     | 28    | 133   | 291   | 246   | 89    | 9     | 0     | 0    |
|                                                                                  | CDH74       | 5870   | 0     | 0     | 5     | 40    | 272   | 968   | 2291  | 1776  | 475   | 43    | 0     | 0    |
|                                                                                  | CDH80       | 1762   | 0     | 0     | 0     | 13    | 77    | 285   | 760   | 528   | 95    | 4     | 0     | 0    |
| Wind                                                                             | WSAvg       | 8.9    | 9.5   | 9.7   | 10.4  | 10.1  | 9.2   | 8.5   | 8.1   | 7.7   | 7.8   | 8.4   | 8.6   | 8.9  |
| Precipitation                                                                    | PrecAvg     | 45.4   | 3.9   | 3.5   | 4.3   | 4.0   | 3.6   | 3.3   | 2.9   | 3.7   | 3.9   | 3.8   | 4.3   | 4.3  |
|                                                                                  | PrecMax     | 67.5   | 11.7  | 7.2   | 8.8   | 12.8  | 8.4   | 11.1  | 8.1   | 10.1  | 7.9   | 15.4  | 11.0  | 10.7 |
|                                                                                  | PrecMin     | 25.4   | 0.5   | 0.4   | 0.6   | 1.5   | 0.7   | 0.2   | 0.4   | 0.7   | 0.9   | 0.4   | 0.4   | 1.0  |
|                                                                                  | PrecStd     | 7.6    | 2.2   | 1.5   | 2.0   | 2.0   | 1.8   | 2.2   | 1.6   | 2.2   | 1.9   | 2.8   | 2.1   | 2.3  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 60.2  | 59.9  | 69.8  | 80.0  | 88.1  | 91.8  | 94.7  | 93.0  | 87.7  | 79.2  | 68.8  | 63.5 |
|                                                                                  |             | MCWB   | 56.5  | 53.2  | 55.4  | 61.9  | 68.7  | 74.1  | 75.7  | 75.7  | 72.9  | 66.4  | 60.4  | 60.1 |
|                                                                                  | 2%          | DB     | 54.7  | 53.8  | 61.1  | 70.7  | 81.1  | 86.9  | 90.5  | 88.6  | 82.3  | 73.6  | 64.5  | 57.8 |
|                                                                                  |             | MCWB   | 50.7  | 47.6  | 50.5  | 56.4  | 65.4  | 71.6  | 74.0  | 73.5  | 69.5  | 65.5  | 58.7  | 53.5 |
|                                                                                  | 5%          | DB     | 49.3  | 49.2  | 55.6  | 65.5  | 76.0  | 83.1  | 87.1  | 85.3  | 79.2  | 70.2  | 61.3  | 53.8 |
|                                                                                  |             | MCWB   | 45.4  | 44.0  | 46.9  | 53.6  | 62.6  | 69.1  | 72.8  | 71.4  | 68.2  | 63.0  | 56.0  | 50.3 |
|                                                                                  | 10%         | DB     | 44.2  | 45.1  | 51.2  | 61.3  | 71.4  | 79.6  | 84.1  | 82.7  | 76.2  | 66.9  | 57.9  | 49.7 |
|                                                                                  |             | MCWB   | 40.3  | 39.9  | 44.4  | 51.0  | 60.1  | 67.6  | 71.3  | 70.4  | 67.2  | 60.5  | 52.9  | 45.2 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 57.7  | 55.8  | 58.7  | 64.0  | 71.4  | 76.7  | 78.6  | 78.3  | 75.3  | 71.2  | 64.0  | 59.9 |
|                                                                                  |             | MCDB   | 59.4  | 58.4  | 66.7  | 72.8  | 84.6  | 87.3  | 89.6  | 89.4  | 81.1  | 75.0  | 66.7  | 62.0 |
|                                                                                  | 2%          | WB     | 51.7  | 49.8  | 53.3  | 59.7  | 67.9  | 74.1  | 76.9  | 76.3  | 73.1  | 67.5  | 60.5  | 54.9 |
|                                                                                  |             | MCDB   | 53.4  | 52.6  | 58.4  | 66.6  | 77.8  | 83.0  | 86.6  | 84.2  | 78.0  | 71.2  | 63.0  | 57.1 |
|                                                                                  | 5%          | WB     | 45.9  | 44.7  | 49.4  | 56.5  | 65.1  | 72.0  | 75.3  | 74.9  | 71.5  | 64.6  | 57.2  | 50.7 |
|                                                                                  |             | MCDB   | 48.7  | 48.1  | 54.2  | 62.5  | 73.2  | 79.4  | 82.7  | 81.1  | 76.2  | 68.7  | 60.1  | 53.1 |
|                                                                                  | 10%         | WB     | 40.8  | 40.4  | 45.3  | 53.4  | 62.2  | 70.1  | 73.8  | 73.3  | 69.7  | 61.7  | 53.8  | 46.0 |
|                                                                                  |             | MCDB   | 43.9  | 44.3  | 50.3  | 58.6  | 68.7  | 76.3  | 80.5  | 78.8  | 74.2  | 66.1  | 57.1  | 49.2 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 14.7  | 15.5  | 16.1  | 17.7  | 17.8  | 17.0  | 16.8  | 16.7  | 16.8  | 17.1  | 15.8  | 14.3 |
|                                                                                  |             | MCDBR  | 20.3  | 20.2  | 22.3  | 24.3  | 24.8  | 21.6  | 19.9  | 19.4  | 19.3  | 20.0  | 18.5  | 19.5 |
|                                                                                  |             | MCWBR  | 18.3  | 16.5  | 14.8  | 13.1  | 12.4  | 10.5  | 8.1   | 8.1   | 10.0  | 12.6  | 14.7  | 17.5 |
|                                                                                  | 5% WB       | MCDBR  | 20.0  | 19.7  | 20.7  | 20.9  | 22.5  | 18.9  | 17.1  | 16.2  | 15.6  | 17.1  | 16.8  | 18.5 |
|                                                                                  |             | MCWBR  | 19.1  | 18.0  | 16.3  | 14.1  | 12.5  | 10.3  | 7.8   | 8.2   | 9.9   | 13.3  | 15.7  | 18.5 |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.299  | 0.308 | 0.332 | 0.372 | 0.414 | 0.460 | 0.483 | 0.441 | 0.373 | 0.338 | 0.311 | 0.298 |      |
|                                                                                  | taud        | 2.499  | 2.480 | 2.463 | 2.381 | 2.296 | 2.206 | 2.150 | 2.255 | 2.420 | 2.530 | 2.564 | 2.560 |      |
|                                                                                  | Ebn at Noon | 272    | 287   | 291   | 285   | 274   | 260   | 253   | 260   | 273   | 272   | 266   | 264   |      |
|                                                                                  | Edh at Noon | 23     | 28    | 31    | 37    | 41    | 45    | 47    | 41    | 33    | 26    | 22    | 20    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 592    | 869   | 1189  | 1530  | 1690  | 1847  | 1907  | 1714  | 1394  | 937   | 642   | 494   |      |
|                                                                                  | RadStd      | 43     | 82    | 84    | 164   | 170   | 184   | 103   | 89    | 102   | 85    | 47    | 49    |      |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                         |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +155  | +92   |
| Regional (70 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +169  | +83   |

Nomenclature: See separate page